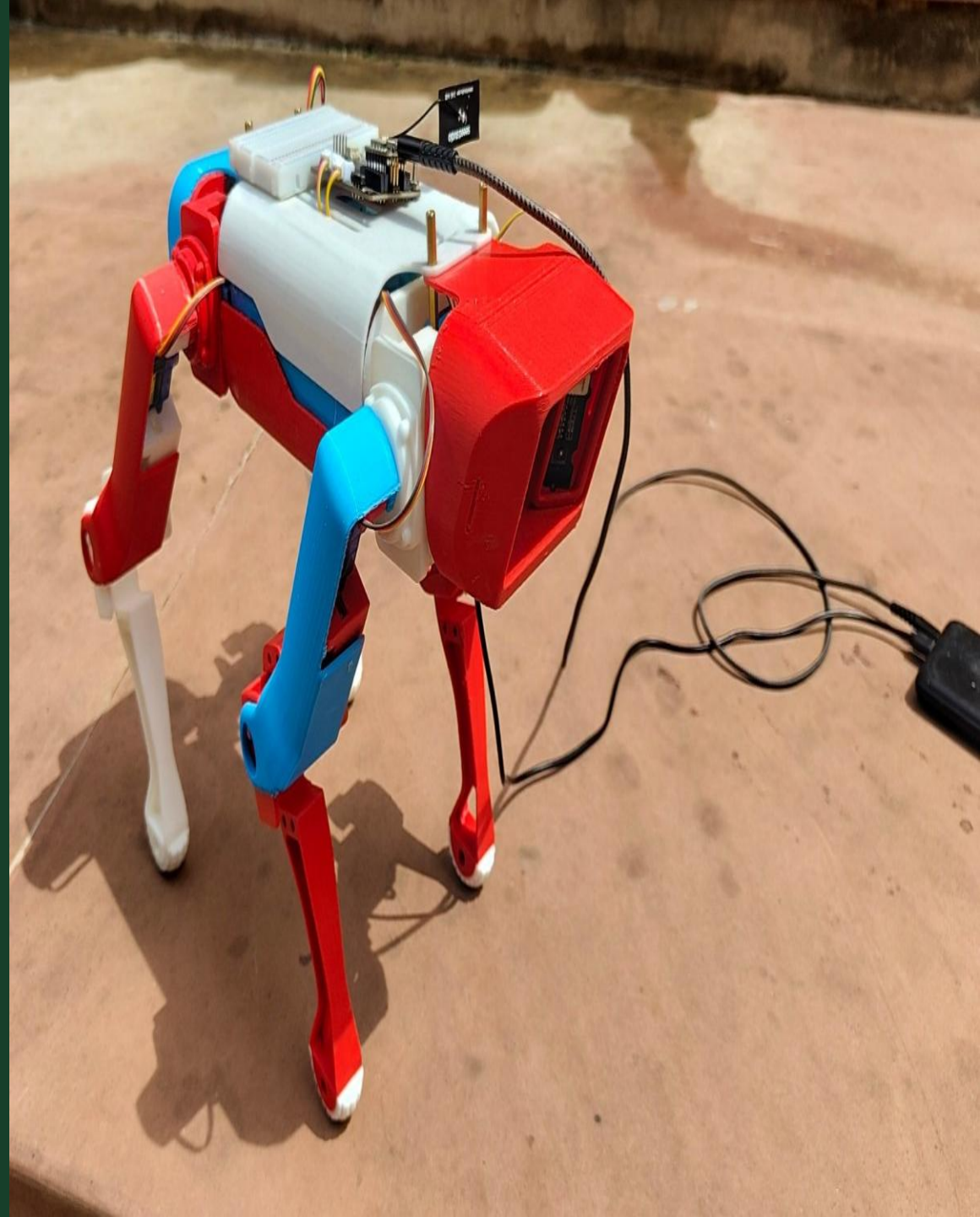
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CAPSTONE PROJECT · FINAL REVIEW

Precision Agriculture Monitoring Quadruped Robot Dog

A 3D-printed quadruped carrying wireless crop-health sensing, with leaf-disease classification and physics-validated locomotion

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How This Review Is Organised

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01	**Domain and Problem**	Crop-loss context, monitoring platforms, 18-paper survey, research gaps, objectives	Slides 3–8
02	**Methodology and Design**	Workflow, design decisions, architecture, data flow, algorithms, locomotion model	Slides 9–19
03	**Completion and Integration**	Scope delivered, assembled robot, integrated dashboard, stable logging, live demo	Slides 20–25
04	**Testing and Results**	Bench tests, classifier validation, hardware actuation, simulated survey, speed limit	Slides 26–33
05	**Innovation and Relevance**	Contributions, cost and feasibility, ethics, practical use, limitations, future work	Slides 34–40
06	**Conclusion and Q&A**	Key takeaways, references, questions	Slides 41–43

01

Domain and Problem



Why crop-level monitoring matters, what existing platforms miss, and what this project set out to do

Crop Disease Is Costly, and Often Seen Too Late

20–40 %

of global crop production is lost to plant pests and diseases every year (FAO)

10–28 %	25–41 %	20–41 %
Wheat	Rice	Maize

Yield losses to pathogens and pests, global estimate (Savary et al., 2019)

Drones see only the canopy top

Short flights; no view of leaf undersides or the air between rows

Wheeled rovers get stuck

Lose traction in mud, soft soil and narrow furrows

Fixed IoT nodes leave gaps

Stationary sensors cover only the spot where they stand

Research robots cost too much

Legged platforms such as ANYmal cost far beyond a small farm's budget

Which Platform Can Monitor Between Crop Rows?

A legged robot can walk between rows and carry sensors to the leaves; the question is whether it can be built cheaply.

Platform	Close-range, leaf-level sensing	Moves on soft soil and in furrows	Covers the whole field	Affordable for small farms
Drone	✗	✓	✓	~
Wheeled or tracked rover	✓	✗	~	~
Fixed IoT sensor nodes	~	–	✗	~
Research quadruped (ANYmal, Cheetah 3)	✓	✓	✓	✗
This project: printed quadruped	✓	✓	✓	✓

✓ suited ~ partly ✗ not suited

What 18 Papers Tell Us

Legged locomotion [1–6]

MIT Cheetah 3, Mini Cheetah, ANYmal, Stanford Doggo; MPC and learned gaits

Gap Custom actuators, costly; none carries a crop-sensing payload

Field robots and drones [7–10]

Reviews of agricultural rovers, drone–ground teaming, farmer interfaces

Gap Wheels fail in mud and furrows; drones cannot see under the canopy

Leaf-disease deep learning [11–15]

CNNs reach 97–99 % on PlantVillage; MobileNetV2 is efficient enough for edge use

Gap Lab images only; closed-set models label anything they see

IoT environmental sensing [16–18]

Layered IoT and sensor networks for microclimate, gases and soil

Gap Stationary nodes leave gaps; nothing carries sensors along rows

Research Gaps and How This Project Answers Them

Gap in the literature	Our response
Low-cost legged robots have not carried an environmental and imaging payload, and their feasibility with hobby servos is rarely quantified	A printed 12-DOF quadruped with two sensing nodes, and a physics model with the real MG996R torque limit
Crop sensing relies on fixed nodes; a mobile, low-cost platform logging gas, dust and temperature along the rows is missing	A wireless sensor node on the robot reporting every 5 s to a logging dashboard
Disease classifiers are closed-set and report a disease for any input, even a wall	A vegetation gate that checks for leaf before the classifier is allowed to answer
Hobby quadruped designs give no guidance on centre of mass or the speed servos allow	Measured: centre of mass 12 mm aft, trot limited to about 0.18 m/s

Objectives and Scope

Mobility

A low-cost printed 12-DOF quadruped whose gait is validated against real servo torque limits, then tested on the robot

Sensing

Gas (TVOC, eCO_2), dust, temperature and range sensing carried on the chassis

Plant-health AI

Leaf-disease classification that first checks a leaf is in view

Live dashboard

Wireless logging, live charts, and leaf images tagged with the readings at capture

In scope

Printed platform, sensing and camera nodes, dashboard and classifier, locomotion design, simulation, first actuation trials

Out of scope

Trials on a working farm, sensor calibration against reference instruments, training a new classifier, GPS mapping

02

Methodology and Design



How the system was designed: the workflow, key technical decisions, architecture, data flow and algorithms

Methodology: Analyse, Revise, Build, Validate

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01 **Analyse**

Torque, power and heat analysis of the Review III plan

02 **Revise**

Move inference to a laptop; split sensing over two ESP32 nodes

03 **Build sensing**

Sensor node and camera firmware, wiring with dividers

04 **Build software**

Flask dashboard, logging, gated classifier

05 **Simulate**

Kinematics and gait in MuJoCo with real servo limits

06 **Validate**

Bench tests, logged sessions, scans, leg actuation

Guiding principle Every design change came from a measurement or analysis: supply and heat limits, servo torque margin, camera interface, and faults found on the bench.

Key Technical Decisions Since Review III

Review III plan	Final system	Reason
Jetson Orin Nano on the robot	XIAO ESP32-S3 and ESP32-CAM nodes; inference on a laptop	Needs 9–19 V and dissipates 7–25 W inside a PLA body; a fall risks an expensive board
IMX219 CSI camera	ESP32-CAM (OV3660), MJPEG over Wi-Fi	The ESP32-S3 has no MIPI-CSI camera interface
ROS 2 and MQTT on the robot	HTTP and JSON to a Flask dashboard	Fewer parts; every device on one local network
Gazebo dynamics model	MuJoCo physics model	The Gazebo stance never stabilised; MuJoCo walks and reports metrics
3S LiPo and onboard regulators	USB power bank for the nodes; separate supply for servos	Safer and simpler for bench work
Walking trials on hardware	Gait in simulation first, then leg actuation on the robot	Simulation showed a centre-of-mass correction is needed

System Architecture

Sensor node

XIAO ESP32-S3 + expansion board

- SGP30 gas · I²C 0x58
- MCP9808 temperature · I²C 0x18
- PPD42 dust · D0
- HC-SR04 × 2 · D6–D9
- OLED, buzzer, button

JSON every 5 s

Laptop base station

Hotspot 192.168.137.1

- Flask dashboard on port 5000
- CSV log of every reading
- Vegetation gate + MobileNetV2
- Relays video to all viewers

MJPEG video, 16–20 fps

Camera node

ESP32-CAM, OV3660 sensor

- MJPEG video on port 81
- Single photo on port 80
- Registers itself at boot

Platform SpotMicro-based printed chassis carrying both nodes · 12 × MG996R servos · gait validated in MuJoCo · any phone on the hotspot can open the dashboard

Data Flow: From Sensor to Tagged Diagnosis

Sensor stream

Sample: gas 1 s, temperature 2 s, dust 30 s

Post JSON to /api/sensors every 5 s

Grade good / moderate / high

Log to daily CSV, update live cards

Camera stream

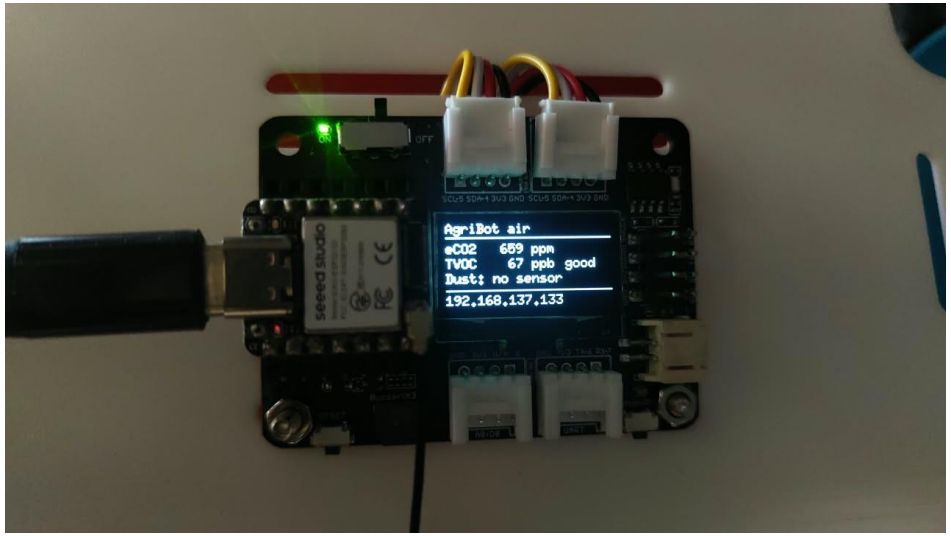
ESP32-CAM streams MJPEG

Dashboard relays frames to every viewer

Scan: vegetation gate, then MobileNetV2

Save image tagged with crop, condition and readings

Sensor Node Hardware



Live readings and IP address on the node's OLED

5 V → 3.3 V safely
Every 5 V sensor output reaches the XIAO through a 1 kΩ / 2 kΩ divider: 5 V becomes 3.33 V, and the PPD42's 4 V becomes 2.67 V.

Sensor	Measures	Interface	Status
SGP30	TVOC, eCO_2	I^2C 0x58	**Live**
MCP9808	Temperature, ±0.25 °C	I^2C 0x18	**Live**
PPD42	Particles > 1 µm	D0, via divider	**Live**
OLED, buzzer, button	Display and alerts	On board	**Live**
HC-SR04 #1	Obstacle, row end	D6 / D7	**Wiring pending**
HC-SR04 #2	Canopy height	D8 / D9	**Wiring pending**

Ultrasonic firmware is written and flashed; the sensors report as absent until connected.

Firmware Algorithms

Dust by low-pulse occupancy

An interrupt times the PPD42's low pulses over 30 s. $C = 1.1r^3 - 3.8r^2 + 520r + 0.62$ particles per 0.01 ft^3

Absent-sensor detection

An unplugged PPD42 read as 1.1 million particles. No edges and a low line now means absent

Temperature-corrected range

$c = 331.3 + 0.606 \cdot T$ m/s from the MCP9808; a fixed 343 m/s is ~3 % off at 35 °C

Crosstalk-free ultrasonics

The two HC-SR04s fire alternately, and each reading is a median of three

Wi-Fi diagnostics

Reports why a link fails; traced −78 dBm to a missing antenna, −26 dBm once fitted

Self-registering camera

The ESP32-CAM sends its stream URL to the dashboard at boot, so a new IP needs no setup

Gated Leaf-Disease Classification

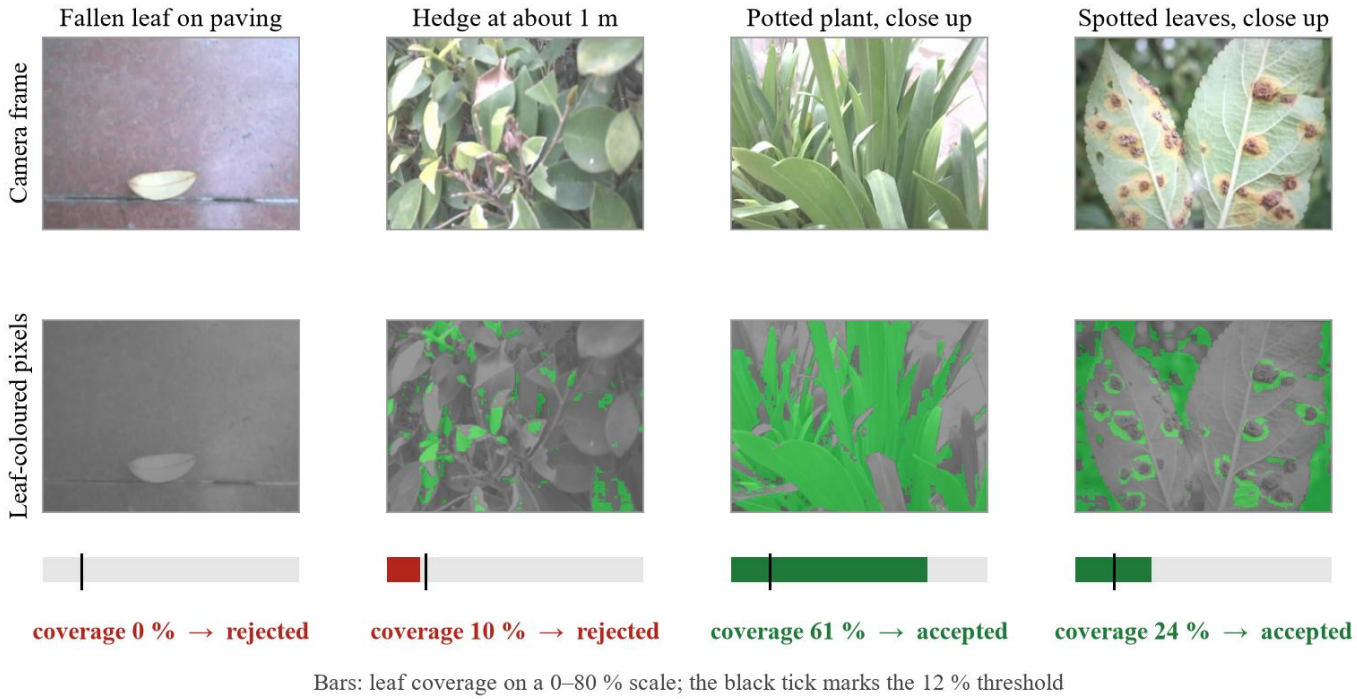
Frame from the ESP32-CAM

Vegetation gate: share of pixels with 38° ≤ H ≤ 165°, S ≥ 0.20, V ≥ 0.12; below 12 % → no leaf

MobileNetV2 trained on PlantVillage: 38 classes, 14 crops

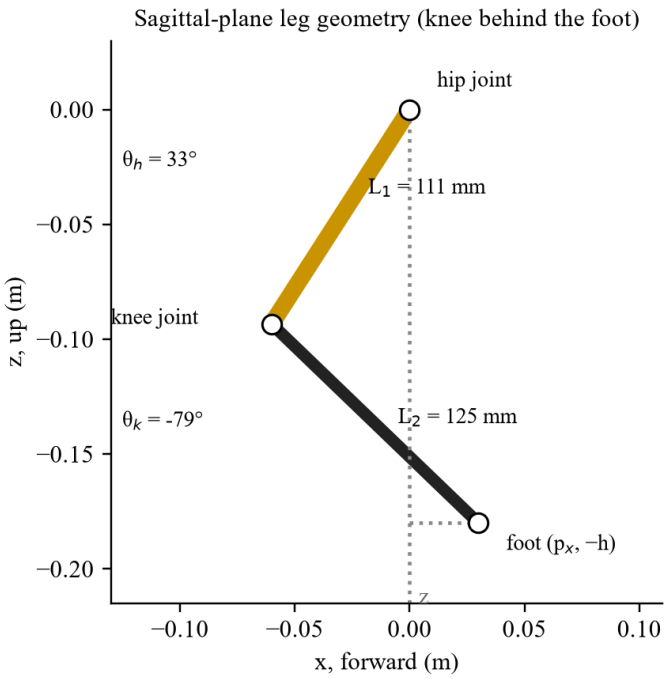
Top probability below 50 % → reported as uncertain

Image saved and tagged with crop, condition and the readings

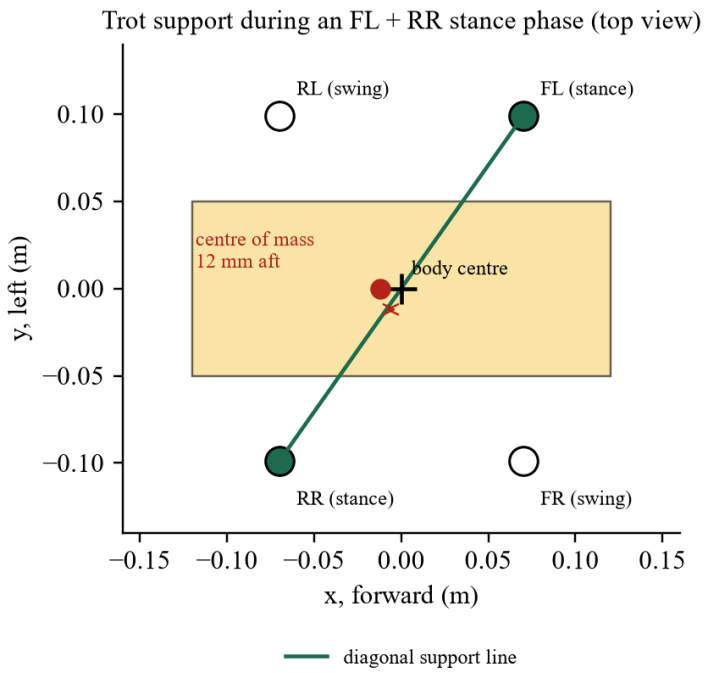


The gate on four outdoor camera frames: leaf-coloured pixels in green

Locomotion Design: Kinematics, Gait, Centre of Mass



Leg geometry: femur 111 mm, tibia 125 mm



Trot support line passes in front of the centre of mass

Closed-form IK

Abduction, hip, knee from the foot target; verified at 1,280 targets, error < 10^{-7} m

Diagonal trot

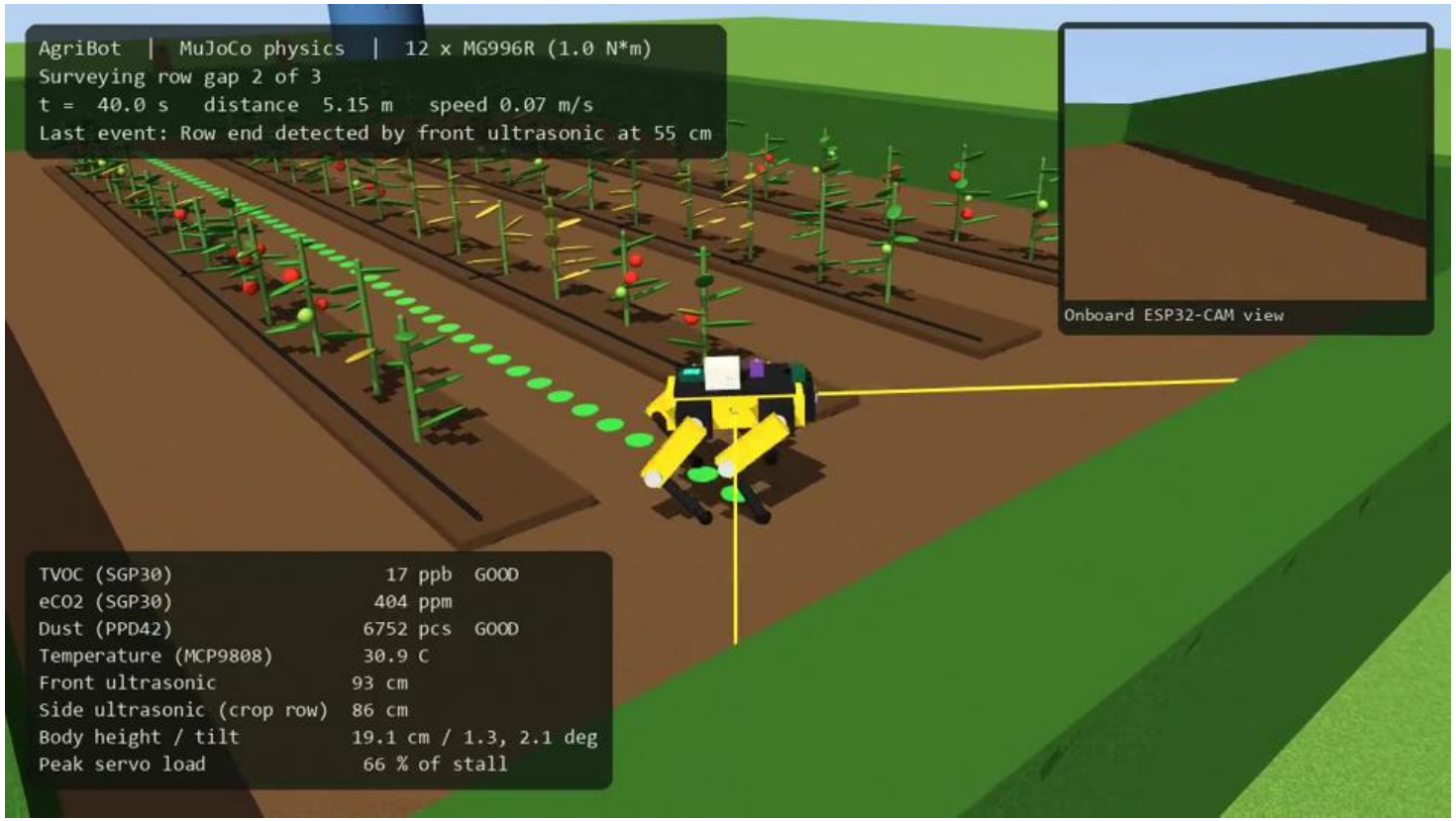
2 Hz, duty 0.6, 30 mm step height, 180 mm body height, roll and pitch levelling

CoM-centred footholds

Feet placed around the measured centre of mass, 12 mm aft of the body centre

Why footholds matter Centred on the body, the trot fell in all 24 gait settings; centred on the measured centre of mass, tilt stayed below 0.7°.

Simulation Model in MuJoCo



Autonomous survey: stand up, trot each row gap, detect the row end, turn, log readings

Measured geometry

140 × 78 mm hip spacing, 111 / 125 mm legs, 1.35 kg

Realistic servos

Position servos capped at 1.0 N·m (MG996R stall 0.92–1.08 N·m)

Modelled field

4 tomato rows, a stressed patch emitting VOCs, a dusty headland

Autonomous mission

Heading control; front range sensor ends each row at 55 cm

Gait chosen by a sweep 24 settings: frequency 2 / 2.5 / 3 Hz × duty 0.5 / 0.6 × stiffness 15 / 30 N·m/rad × levelling gain 0 / 0.5

Technology Stack

Hardware

XIAO ESP32-S3 + expansion board · ESP32-CAM (OV3660) · SGP30, MCP9808, PPD42, HC-SR04 × 2 · 12 × MG996R

Firmware

Arduino-ESP32 core 3.3 · Adafruit SGP30 · U8g2 display · esp_http_server for MJPEG

Dashboard and AI

Python · Flask · PyTorch · Hugging Face Transformers · MobileNetV2 (PlantVillage)

Simulation

MuJoCo 3 · NumPy · Matplotlib · headless video and metrics

Mechanical

SpotMicro base (CC-BY) · PLA+ and TPU · Python generators for the custom mounts

Collaboration

Git and GitHub: firmware, dashboard, simulation and print files in one repository

03

Completion and Integration



What was delivered, how the parts work together, and evidence of stable operation, followed by the live demo

Scope Delivered

4 / 6

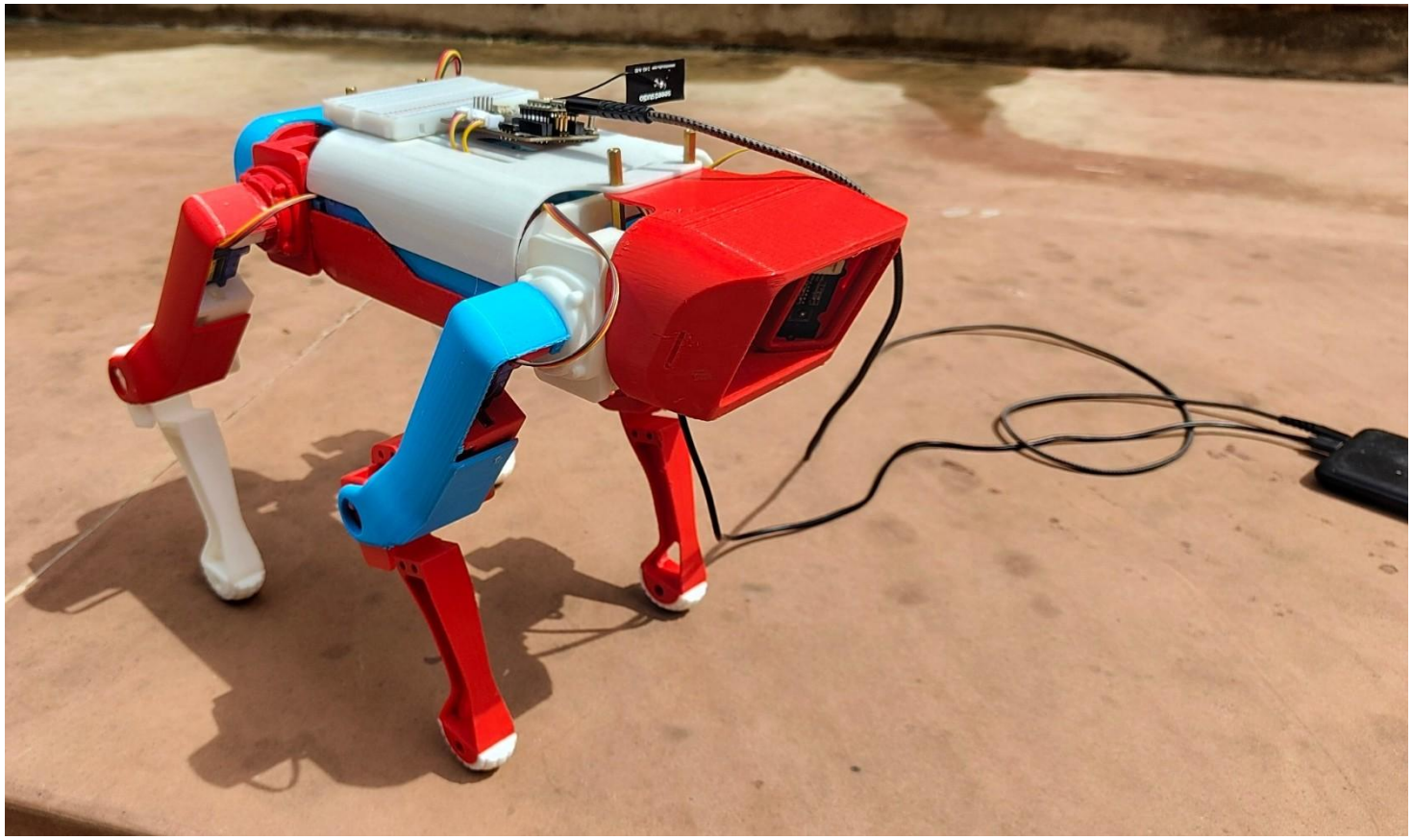
objectives fully achieved

2 / 6

partly achieved, with a clear next step

Objective	Status	Evidence
Print and adapt the chassis; measure its geometry	**Achieved**	35 parts printed; four custom parts; measured geometry
Wireless sensing payload, robust to faults	**Partly**	Gas, dust, temperature live; ultrasonic wiring pending
Camera streaming and gated classification	**Achieved**	16–20 fps; gate rejects non-leaf frames
Live dashboard with logging and tagging	**Achieved**	Logged sessions; tagged captures with readings
Verified kinematics; gait feasible in simulation	**Achieved**	IK error < 10^{-7} m; full survey, no fall
Walking on the physical robot	**Partly**	Legs actuate and it stands; no steps yet

The Assembled Robot



The assembled robot standing outdoors

35 printed parts

SpotMicro base adapted for the payload; ~700 g PLA+ with TPU feet; ₹5,200 to print

Sensor node on the top cover

XIAO, breadboard and sensors, with the external antenna on a standoff

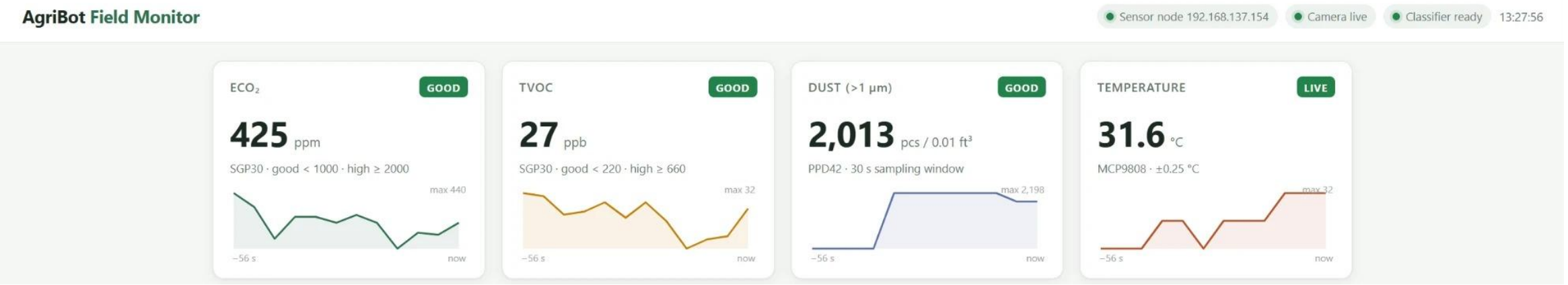
Camera in the head

ESP32-CAM looks out through the printed nose

No battery on board

Payload runs from a USB power bank beside the robot

Integrated Dashboard



Dashboard on 18 October 2026: live eCO_2, TVOC, dust and temperature from the sensor node

Status bands

TVOC 220 / 660 ppb (German UBA guideline); eCO_2 1,000 / 2,000 ppm; node marked offline after 15 s of silence

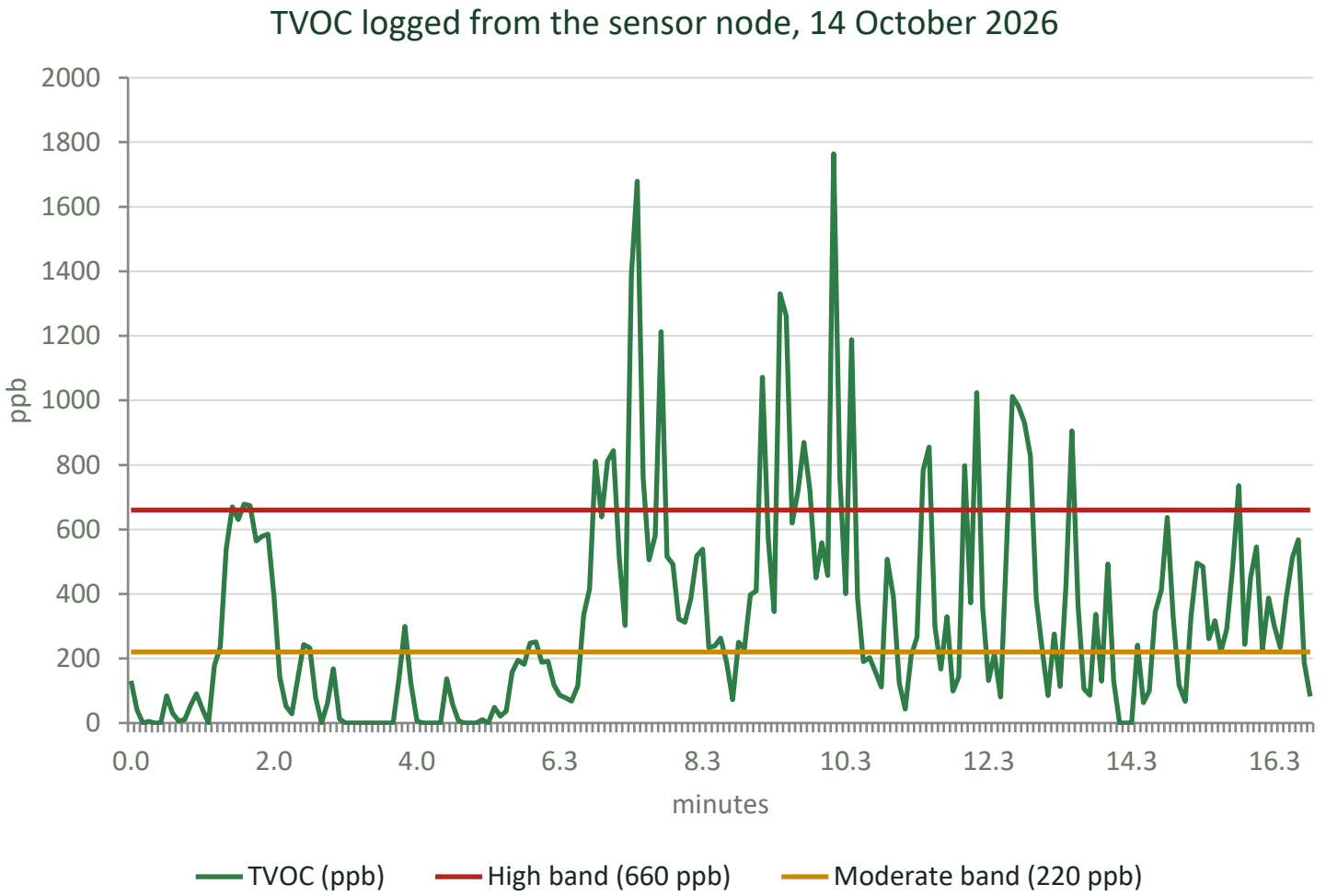
Logging

Every reading to a daily CSV; every scan saved as a tagged image with the readings at capture

Camera and diagnosis

Live MJPEG video, one-click scan or image upload, top three classes with confidence

Stable Operation over Wi-Fi



321 + 199

readings in two sessions (28 and 17 min), no records lost

1,764 ppb

TVOC peak during exposure, back to baseline within tens of seconds

−26 to −60

dBm Wi-Fi signal throughout, ample for 5 s telemetry

Live Demonstration

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1 **Dashboard live**

Open http://192.168.137.1:5000; status pills show the sensor node, camera and classifier

2 **Breath test**

Breathe on the SGP30: eCO_2 and TVOC rise within seconds and recover

3 **Touch test**

Hold the MCP9808: temperature climbs about 3–4 °C

4 **Leaf scan**

Show a leaf to the camera, then an empty scene: diagnosis versus no leaf in view

5 **Simulation**

Run the MuJoCo survey: rows, turns, TVOC trail over the stressed patch

6 **Robot**

Power the servos: the robot stands and moves its legs

04

Testing and Results



Test procedures and measured results on the hardware, then the simulation metrics and their interpretation

Bench Validation of the Hardware

Test procedure	Measured result	What it shows
Breathe on the SGP30	eCO_2 400 → 746 ppm, TVOC 0 → 210 ppb within 5 s; back to baseline	Gas sensing responds and recovers
Finger on the MCP9808	31.5 → 35.0 °C in about 10 s	Temperature tracks a real change
PPD42 in room air	0–5,065 particles / 0.01 ft^3 per 30 s window	Dust sensing live
PPD42 unplugged	Reported as absent instead of 1.1 million	No false alarms from wiring faults
Wi-Fi link of the XIAO	−78 dBm with time-outs → −26 dBm with antenna	Diagnostics found the fault
Camera video	≈0.5 → 16–20 fps after moving to MJPEG	Live view usable for scanning
Servos, robot on its back	All four legs follow commanded angles	Wiring and servo control work
Servos, robot standing	Holds a stance and shifts its legs, slowly	Torque margin is thin, as predicted

Classifier Validation: The Gate Prevents False Diagnoses

Without the gate

82–90 %

confidence for “citrus greening” on a photo of an empty room

With the gate

Rejected

0–0.1 % leaf coverage measured: “no leaf in view”, no diagnosis shown

12 / 12

PlantVillage leaves accepted by the gate (27–76 % coverage)

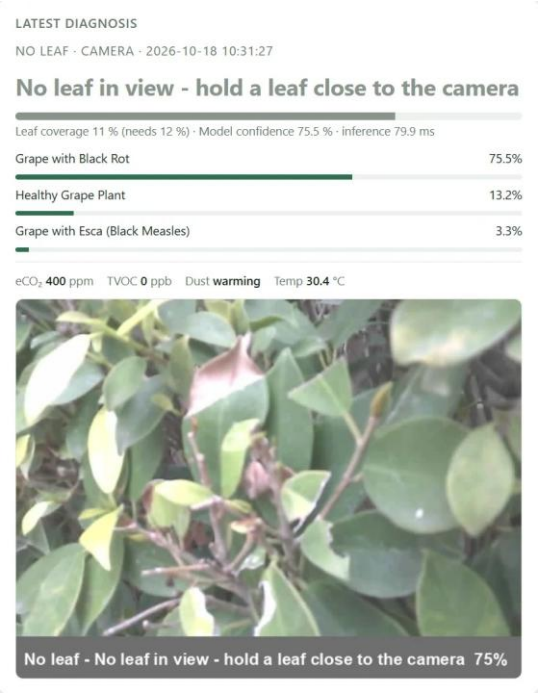
8 / 10

spot-check images classified correctly above 92 %; 1 uncertain, 1 wrong

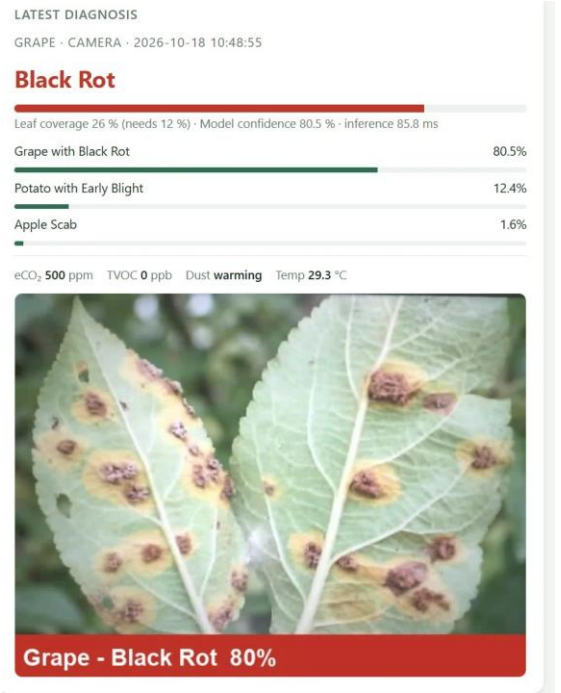
30–80 ms

inference per image on the laptop CPU

Live Scans Outdoors with the Robot's Camera







Hedge at ~1 m: no leaf in view

11 % coverage, so the model's 75.5 % “black rot” is withheld

Potted plant: low confidence

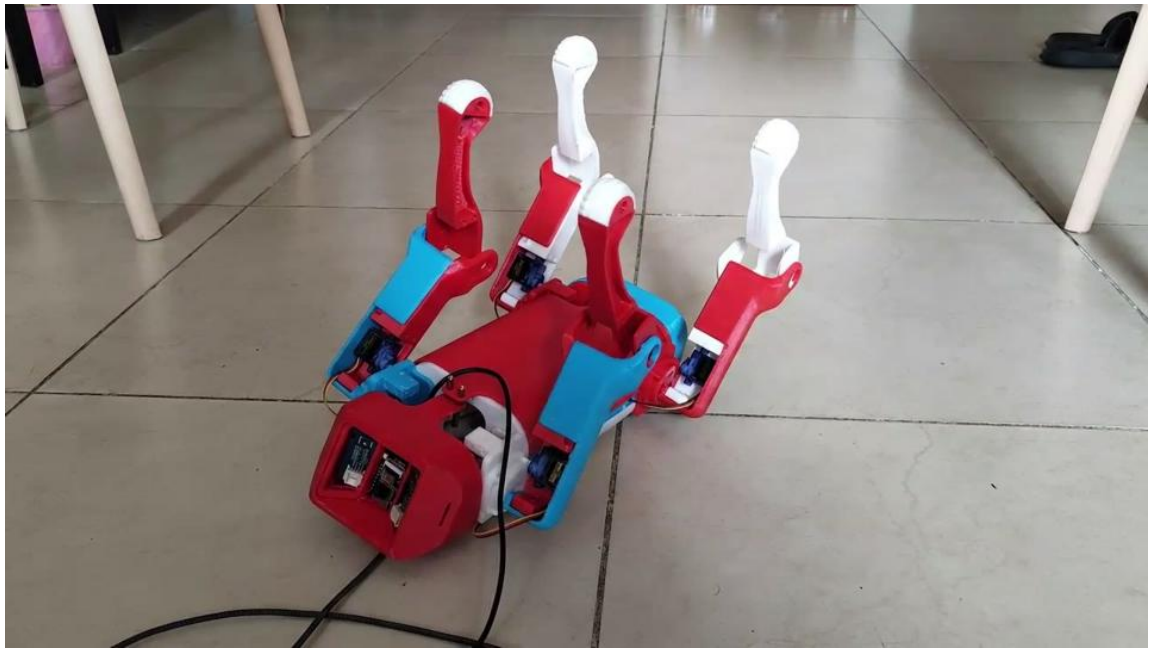
60 % coverage; “healthy corn” at 53.4 %, just above the 50 % threshold

Spotted leaves: flagged

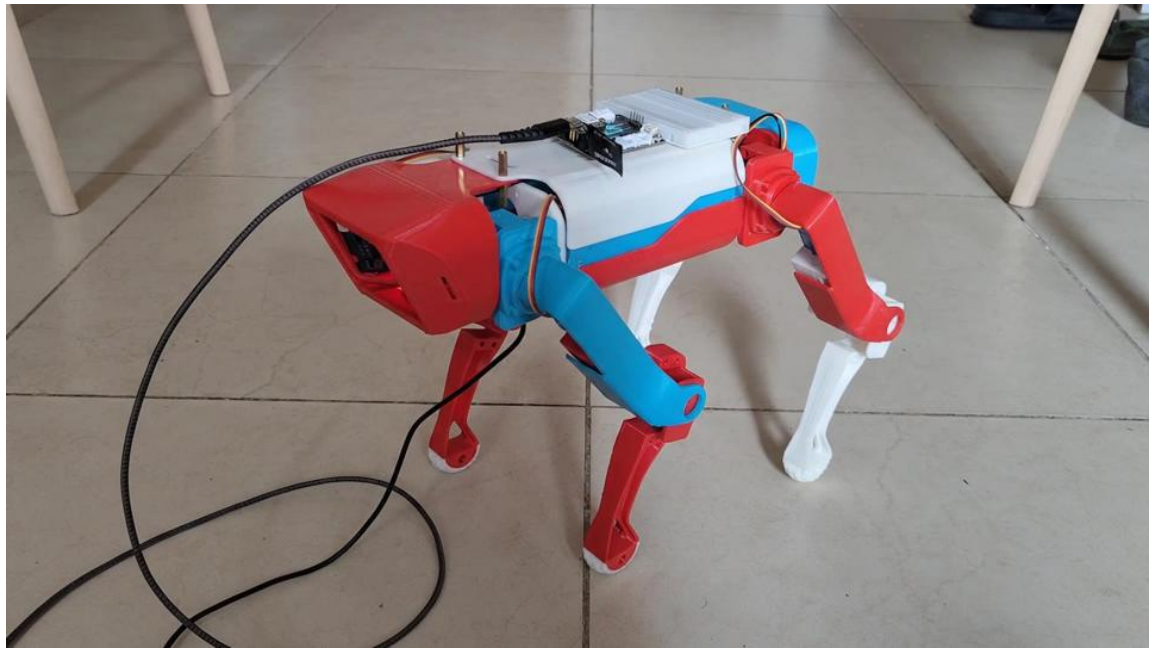
26 % coverage; “grape black rot” at 80.5 %, readings saved with the image

Washed-out camera colours put some real foliage near the 12 % threshold; holding the leaf close fixes it.

Actuation Trials on the Physical Robot



On its back: all four legs follow their commanded angles



Standing: supports its weight and shifts its legs, but weakly

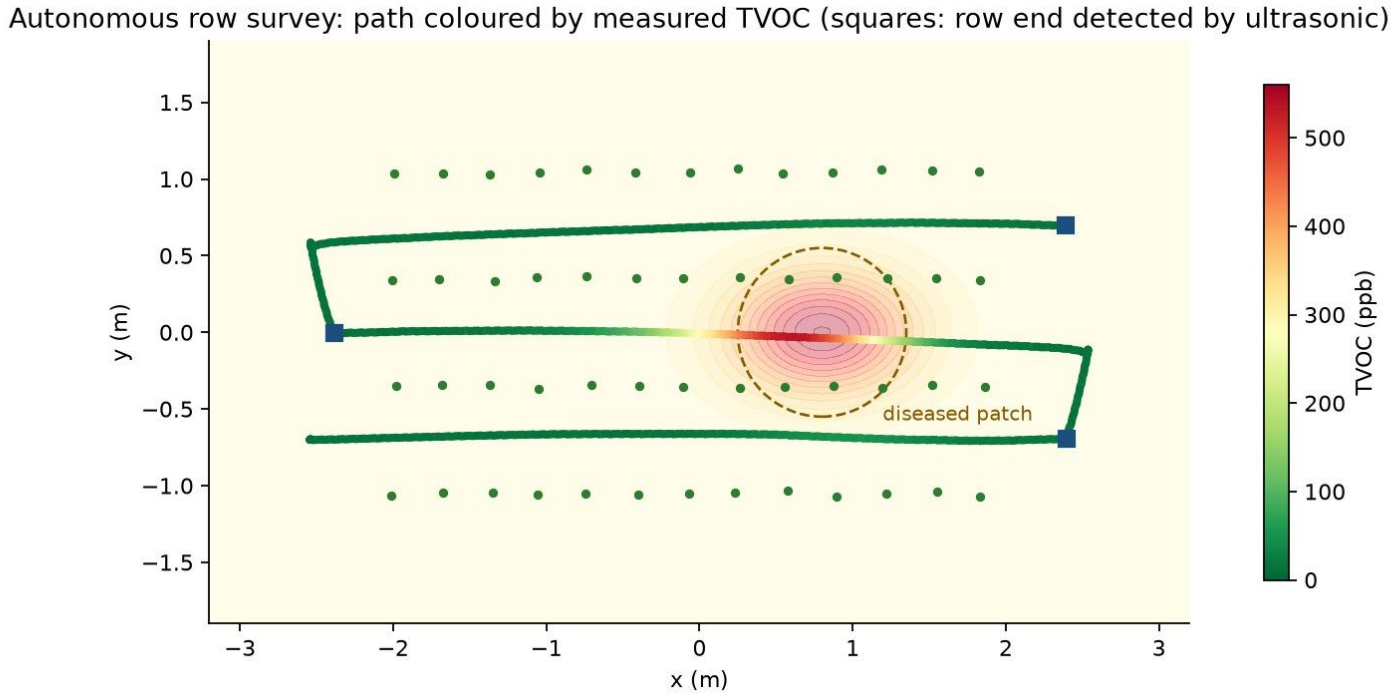
Why it is weak

Stance knee torque is 36–52 % of a genuine MG996R's rating and rises when a leg lifts; copies deliver less, and a sagging supply cuts torque further.

Next step

Dedicated 6 V servo supply, CoM-centred footholds and straighter legs (a taller stance); stronger servos if the margin is still thin.

Simulated Autonomous Field Survey



Route coloured by measured TVOC over the modelled VOC field; squares mark row ends detected by the front ultrasonic sensor

16.7 m

in 129 s: all three row gaps, no falls

3 / 3

row ends detected by the ultrasonic sensor

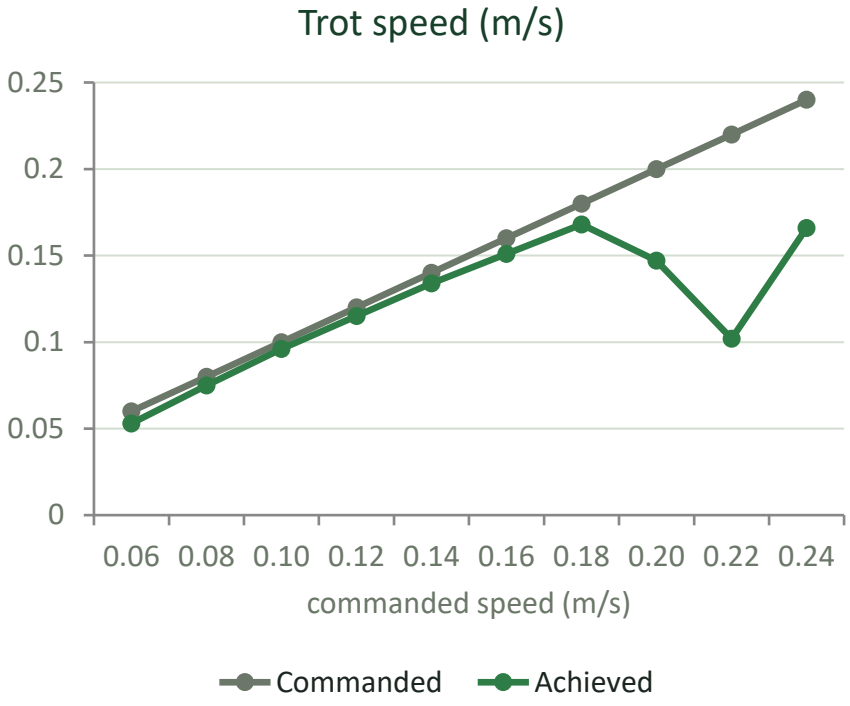
≤ 2.7°

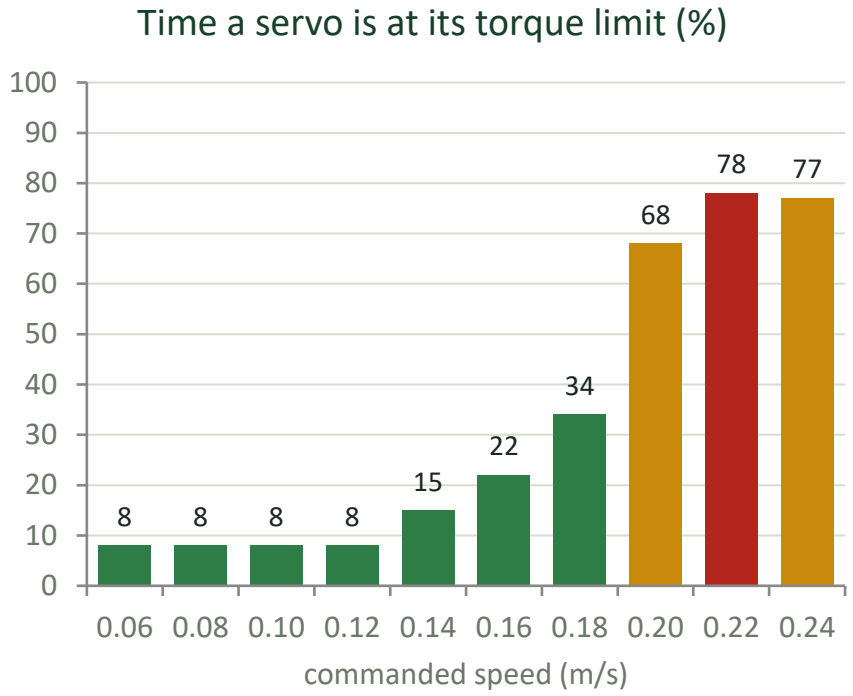
body tilt; a servo at its limit 1.3 % of the time

535 ppb

TVOC peak, 0.2 m from the source (sensor lag)

Gait Stability and the Servo Speed Limit





0.18 m/s

usable ceiling with MG996R servos; tracks the command within 7 % from 0.08 m/s up

Centre of mass 12 mm aft With footholds centred on the body, all 24 gait settings fell within 0.8–3.0 s. Centred on the measured centre of mass, the trot held tilt below 0.7°.

Outcome Against Review III Expectations

Review III expectation	Outcome	Interpretation
Onboard inference on a Jetson	Inference on a laptop over Wi-Fi	Cheaper and safer; depends on the network link
Above 90 % disease accuracy (literature figure)	Functional demonstration; field accuracy not measured	Lab-trained models are expected to drop on field images
Static torque margin is thin	Confirmed in simulation (0.18 m/s ceiling) and on hardware	Analysis, simulation and robot agree
Walking robot with sensing	Robot stands and moves its legs; walking shown in simulation	Hardware walking is the next phase
Environmental sensing payload	Gas, dust, temperature live; ultrasonic firmware ready	Sensing payload essentially complete

05

Innovation and Relevance



What is new, whether it is feasible and affordable, how it is used responsibly, and where it goes next

Contributions

Legged platform with a crop-sensing payload

A printed 12-DOF quadruped carrying two Wi-Fi sensing nodes for gas, dust, temperature, range and video

Vegetation gate for a closed-set classifier

A simple, explainable leaf check that stops confident disease reports for frames without a plant

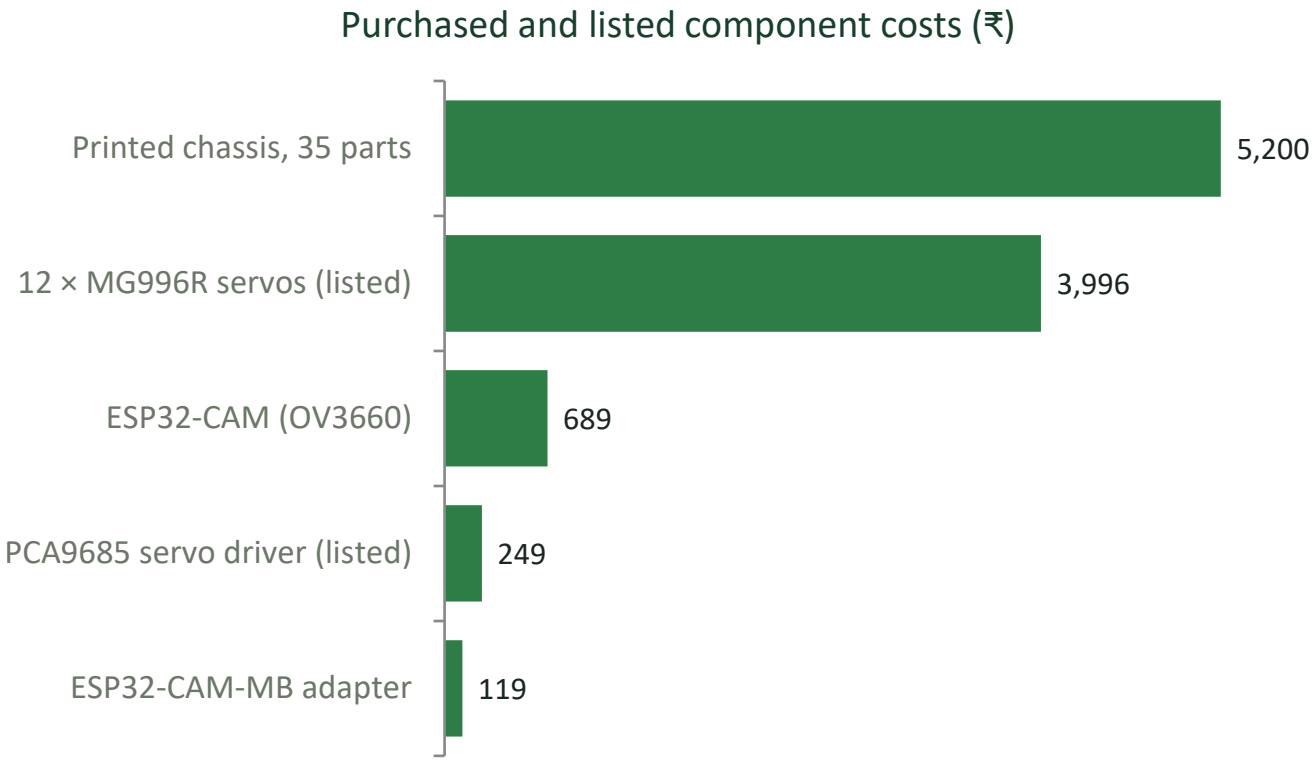
Locomotion validated with real servo limits

A MuJoCo model from measured geometry, torque capped at MG996R stall, completing an autonomous row survey

Design rules for printed quadrupeds

Centre footholds on the measured centre of mass (12 mm aft); expect about 0.18 m/s from MG996R servos

Cost and Feasibility



The XIAO, sensors, resistors and cables were already available to the team and are not costed.

≈ ₹10,250

total purchased and listed cost of the platform

Reported cost of research quadrupeds

Platform	Reported cost
Stanford Doggo [2]	< US$3,000
MIT Cheetah 3 [1]	~US$50,000+
ANYmal [4]	US$100,000+

Ethical and Responsible Use

Data stays local

All readings and images stay on the laptop's own hotspot; no cloud service, no personal data, camera aimed at crops

No false certainty

Gate and 50 % threshold suppress unreliable labels; results are indications for an agronomist, not a spray decision

Electrical and physical safety

Low-voltage USB payload, no LiPo on board, slow-moving robot, separate servo supply

Honest sensor claims

eCO_2 is estimated and the dust sensor uncalibrated; used for trends and alerts, not regulatory values

Open source and attribution

SpotMicro design credited under CC-BY; our code and print files are public

Built for small farms

Low-cost parts and no internet needed, so the approach is within reach of smallholders

Practical Applicability

Row-by-row scouting

Walk the rows, log gas, dust and temperature, and scan suspicious leaves on the way

Early-warning hotspots

Rising VOC readings along a row point to stressed plants before symptoms spread

Nurseries and greenhouses

Repeated inspection of seedlings in narrow aisles where wheels struggle

Records for agronomists

Tagged images with the conditions at capture give an evidence trail for advice

What a farm needs to run it

- A laptop with Wi-Fi hotspot
- The robot with its two nodes
- A USB power bank
- Any phone to view the dashboard
- No internet, no cloud account

Dashboard address
http://192.168.137.1:5000

Limitations

Limitation	Effect	How it is handled
Real servos lack torque reserve	Robot stands and moves its legs but does not step	Dedicated supply, CoM footholds, taller stance, stronger servos
Classifier trained on lab photos, 14 crops	Lower accuracy on field images; any plant passes the gate	Gate and 50 % threshold; fine-tuning on field images
Washed-out ESP32-CAM colours	Some real foliage falls below the 12 % threshold	Hold the leaf close; tune the threshold to this camera
eCO_2 estimated; PPD42 uncalibrated	Relative readings, not regulatory values	Used for trends and alerts
Simulated fields are modelled	Simulation shows the method, not field data	Hardware sensors validated on the bench

Future Work

01	**Walking on hardware**	Dedicated 6 V servo supply; measure torque and voltage under load; CoM footholds and a taller stance; step in place, then walk below 0.18 m/s
02	**Complete the sensor set**	Wire both ultrasonic sensors; add a soil-moisture probe; calibrate gas and dust against reference instruments
03	**Act on measurements**	Grove relay to switch an irrigation pump or raise a gas alarm from the dashboard
04	**Field-ready classification**	Fine-tune on images from the robot's camera; replace the colour gate with a learned open-set detector
05	**Mapping and autonomy**	Compensate for the SGP30's lag; log positions to build field maps; battery for untethered runs

Conclusion

VIT CHENNAI

Sensing works end to end

Two wireless nodes, a live dashboard with logging, and leaf-disease classification that refuses to guess without a leaf

Locomotion validated in physics

A full autonomous row survey without a fall, with servos capped at their real torque

The printed robot moves

All twelve servos drive the legs and it stands, with too little torque reserve to step, as predicted

Design rules for the next build

Centre footholds on the centre of mass; expect about 0.18 m/s from MG996R servos

Repository github.com/AdithyaSM31/Precision-Agriculture-Monitoring-Quadruped-Robot-Dog

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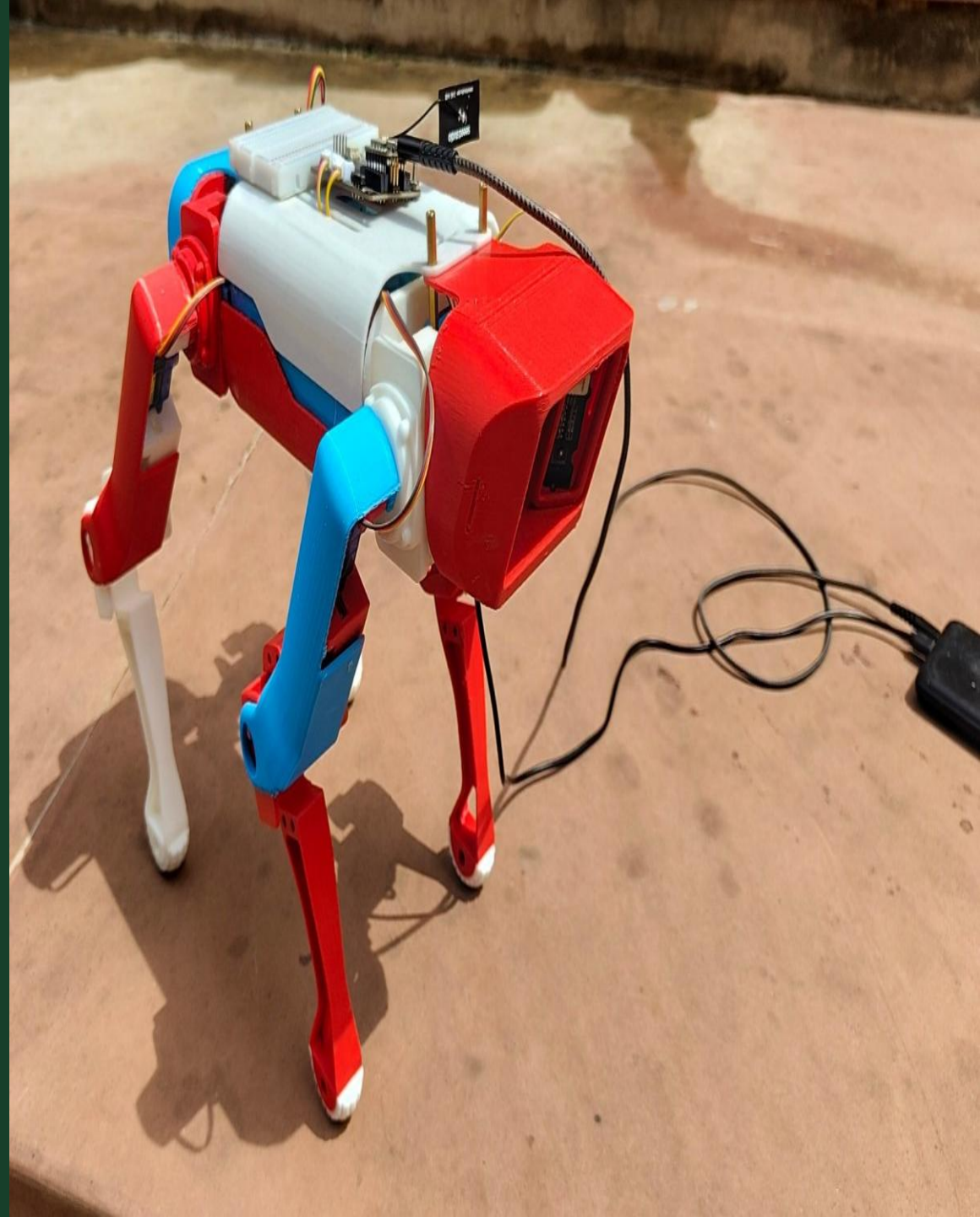
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Thank you

Questions and live demonstration

Adithya Sankar Menon · Siddhesh Pushkraj Shinde · Rachit Gupta
Guide: Dr. Lakshmanan K



Backup: Literature Review, Papers 1–6

#	Authors and year	Title	Key contribution	Research gap
1	Bledt et al. (2018)	MIT Cheetah 3: Design and Control of a Robust, Dynamic Quadruped Robot	Designed high-torque density proprioceptive actuators and Model Predictive Control (MPC) for dynamic walking over rough terrain without vision.	High cost (~$50k+), complex custom planetary gearing, no environmental/agricultural sensing payload.
2	Kau et al. (2019)	Stanford Doggo: An Open-Source, Quasi-Direct-Drive Quadruped	Developed an open-source 3D-printed quadruped platform using quasi-direct drive brushless motors for dynamic bounding.	Optimized purely for acrobatic jumps/speed; lacks on-board compute for perception and has <10 min battery life.
3	Katz et al. (2019)	Mini Cheetah: A Hardware Design for Extremity Locomotion	Modular 3-DOF leg design using low-cost backdrivable BLDC motors enabling agile omnidirectional gaits.	Focuses strictly on mechanical agility in laboratory environments; no autonomous navigation or edge AI pipeline.
4	Hutter et al. (2016)	ANYmal - A Unique Quadruped Robot for Autonomous Operation in Harsh Environments	Industrial-grade quadruped with series elastic actuators (SEAs) and IP67 ruggedized body for outdoor inspection.	Prohibitively expensive ($100k+), heavy industrial form-factor unsuitable for smallholder precision farming.
5	Ding et al. (2021)	Learning to Walk on Uneven Terrain for Quadrupedal Robots using Reinforcement Learning	Trained neural network locomotion policies in simulation (Isaac Gym) with robust sim-to-real transfer over rugged slopes.	High computational training overhead; policy does not integrate multi-sensor agronomic payloads or path tracking.
6	Bellicoso et al. (2018)	Dynamic Locomotion Through Online Nonlinear Model Predictive Control for Quadrupedal Robots	Real-time ZMP and CoM trajectory optimization enabling adaptive crawl and trot gaits over unmodeled obstacles.	Requires high-end x86 computer boards; difficult to run concurrently with edge vision models on low-power platforms.

Backup: Literature Review, Papers 7–12

#	Authors and year	Title	Key contribution	Research gap
7	Fehlmann et al. (2023)	Precision Agriculture Using Autonomous Ground Robots: A Review	Systematic review of wheeled and tracked agricultural rovers for weed control, phenotyping, and soil sampling.	Identifies severe mobility failure of wheeled rovers in wet mud, soft soil, and high-crop canopy furrows.
8	Roldán et al. (2018)	Mini-UAVs and Mobile Robots for Precision Agriculture: A Review of Multi-Robot Systems	Analyzed complementary roles of aerial drones (macro-surveys) and ground robots (micro-inspection).	Highlighted drone limitations: 20-min flight time, inability to inspect leaf undersides, and blind spots beneath crop canopies.
9	Bogue, R. (2016)	Robots in Agriculture: State of the Art and Future Directions	Surveyed commercial adoption of agricultural robotics, highlighting high ROI in crop health monitoring.	Commercial units are bulky, heavy, cause soil compaction, and require flat structured greenhouse tracks.
10	Vasconez et al. (2020)	Human–Robot Interaction in Agriculture: A Survey and Review	Investigated telemetry, UX dashboards, and safety mechanisms for farmer-robot interfaces in open fields.	Found most agricultural robotic interfaces are academic prototypes lacking real-time edge telemetry and alert maps.
11	Mohanty et al. (2016)	Using Deep Learning for Image-Based Plant Disease Detection	Trained AlexNet and GoogLeNet on 54,306 images across 26 diseases with 99.35% classification accuracy.	Tested only on static laboratory-controlled leaf images with uniform backgrounds; lacks edge device deployment.
12	Ferentinos, K. P. (2018)	Deep Learning Models for Plant Disease Detection and Diagnosis	Evaluated VGG, ResNet, and AlexNet on 87,848 leaf images with real-world noise variations (99.5% accuracy).	High parameter count (VGG/ResNet) causes severe latency and memory bottlenecks on embedded single-board computers.

Backup: Literature Review, Papers 13–18

#	Authors and year	Title	Key contribution	Research gap
13	Too et al. (2019)	A Comparative Study of Fine-Tuning Deep Learning Architectures for Plant Disease Identification	Benchmarked MobileNetV2, DenseNet-121, and InceptionV3; proved MobileNetV2 achieves 97.4% accuracy with 10× fewer FLOPs.	Did not perform hardware-specific quantization (TensorRT / ONNX) or test real-time video stream ingestion.
14	Chen et al. (2020)	Using Deep Transfer Learning for Image-Based Plant Disease Identification on Embedded Devices	Deployed lightweight CNN on Raspberry Pi 3B+ using INT8 quantization, achieving 12 FPS at 92.1% accuracy.	Insufficient FPS for continuous mobile robot navigation; no integration with robotic gait or sensor fusion.
15	Saleem et al. (2019)	Plant Disease Detection and Classification by Deep Learning: A Review	Comprehensive meta-analysis of deep learning architectures, dataset biases, and real-time field challenges.	Concluded that field-ready solutions require mobile ground robots that bring cameras within 0.5m of plant leaves.
16	Ray, P. P. (2017)	Internet of Things for Smart Agriculture: Technologies, Practices and Future Direction	Proposed layered IoT architecture for soil moisture, microclimate, and particulate monitoring in precision farming.	Relied exclusively on stationary sensor nodes (motes), creating spatial coverage gaps across large farm tracts.
17	Elijah et al. (2018)	An Overview of Internet of Things (IoT) and Data Analytics in Smart Agriculture	Investigated wireless sensor networks (WSNs) and cloud data pipelines for environmental anomaly detection.	High infrastructure cost of deploying hundreds of stationary nodes; vulnerable to field vandalism and weather damage.
18	Bacco et al. (2019)	Smart Farming: Technologies, Applications and Challenges in Precision Agriculture	Explored multi-sensor data fusion (temperature, humidity, organic gases, optical imagery) for soil health profiling.	Lacked mobile autonomous platforms to transport multi-sensor payloads dynamically throughout crop rows.